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TECHNICAL SERVICE & PARTS MANUAL

Single Piston Floating Caliper Assembly

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1. System Overview

The front disc brake system provides controlled braking deceleration by converting hydraulic pressure into mechanical clamping force against the brake rotor.

1.1 Principle of Operation

When the rider applies the front brake lever, hydraulic fluid from the master cylinder forces the caliper piston (Item 8) outward. This movement presses the inner brake pad against the rotating disc.

As hydraulic pressure increases, the floating caliper body (Item 10) slides along the guide pins (Items 1 and 2), pulling the outer brake pad against the opposite surface of the disc. This dual clamping action generates friction to reduce wheel speed.

1.2 Key Service Specifications

- **Brake fluid specification:** DOT 4
- **Standard pad friction thickness:** 4.5 mm
- **Service wear limit:** 1.5 mm

2. Brake Pad Replacement Procedure

Prerequisites & Safety:

- Park the vehicle on a level surface and secure the center stand.
- **Required Tools:** 12 mm socket wrench, 8 mm hex key, vernier caliper, torque wrench.

> **CAUTION:** Brake fluid damages painted surfaces. Wipe spills immediately with a clean, dry cloth.

Step-by-Step Removal & Installation:

1. Loosen the guide pin (Item 1) and lock pin (Item 2) by two turns using an 8 mm hex key. *(Do not remove the pins completely at this stage.)*
2. Remove the caliper mounting bolts securing the caliper body (Item 10) to the fork bracket.
3. Slide the caliper body (Item 10) backward away from the brake rotor. *(The caliper is now free from the wheel assembly.)*
4. Unscrew and remove guide pin (Item 1) and lock pin (Item 2).

5. Remove the inner pad with wear indicator (Item 11) and the outer pad (Item 12).
6. Measure the friction lining thickness of both pads using a vernier caliper. *(If lining thickness is less than 1.5 mm, replace both pads as a complete set.)*
7. Clean the caliper bracket, pad clips, and guide pins with approved brake cleaner.
8. Install the new inner pad (Item 11) and outer pad (Item 12) into the caliper support (Item 4).
9. Tighten the guide pin (Item 1) to **27 Nm** and lock pin (Item 2) to **22 Nm** using a torque wrench.

Post-requisite: Pump the brake lever several times to seat the pads against the rotor before operating the vehicle.

3. Illustrated Parts Catalog (IPC)

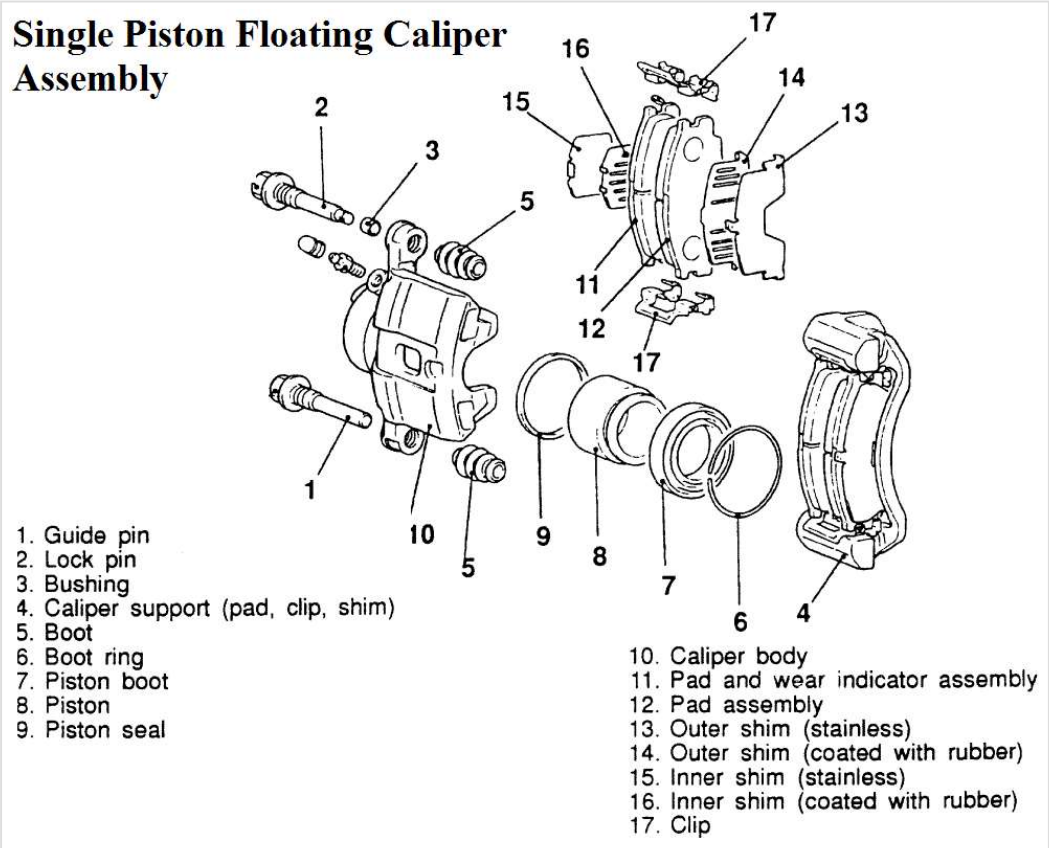


Figure 1: Single Piston Floating Caliper - Exploded View

Item No.	Part Number	Description	Qty	Remarks / Torque
1	5120-GP-01	Guide Pin	1	Torque: 27 Nm
2	5120-LP-02	Lock Pin	1	Torque: 22 Nm
4	5120-CS-04	Caliper Support	1	Carrier bracket
8	5128-CP-32	Piston	1	Diameter: 32 mm

Item No.	Part Number	Description	Qty	Remarks / Torque
9	5129-PS-32	Piston Seal	1	Replace upon overhaul
10	5120-CB-10	Caliper Body	1	Cast iron/alloy
11	5122-PD-11	Pad & Wear Indicator Assembly	1	Inner pad
12	5122-PD-12	Pad Assembly	1	Outer pad